30) Let $\bar{v}$ be the center of mass of a system of point masses located at $v_1, \ldots, v_k$ as in Exercise 29. Is $\bar{v}$ in Span$\{v_1, \ldots, v_k\}$? Explain.

$\boxed{\text{Solution:}}$ From Ex-29, the center of mass of the system is

$$\bar{v} = \frac{1}{m}[m_1 v_1 + \ldots + m_k v_k] = \left(\frac{m_1}{m}\right)v_1 + \ldots + \left(\frac{m_k}{m}\right)v_k.$$

It's easy to see that $\forall i, i \in \mathbb{N}, 1 \leq i \leq k, \left(\frac{m_i}{m}\right) \in \mathbb{R}$.

$\therefore \bar{v}$ is a linear combination of the vectors $v_1, \ldots, v_k$.

By Defn, $\bar{v}$ is in Span$\{v_1, \ldots, v_k\}$

31) A thin triangular plate of uniform density and thickness has vertices at $v_1 = (0,1)$, $v_2 = (8,1)$, and $v_3 = (2,4)$, as in the figure below, and the mass of the plate is 3gm.

a) Find the (x_1, x_2) coordinates of the center of mass of the plate. This "balance point" of the plate coincides with the center of mass of a system consisting of three 1-gram point masses located at the vertices of the plate.

$\boxed{\text{Solution:}}$ $v_1 = (0,1)$ $v_2 = (8,1)$ $v_3 = (2,4)$ $m_1 = m_2 = m_3 = 1$ gm

$m = 3m_1 = 3$gm

Center of mass: $\frac{1}{3}\left[1(0,1) + 1(8,1) + 1(2,4)\right] = \frac{1}{3}\left[(1\cdot 0, 1\cdot 1) + (1\cdot 8, 1\cdot 1) + (1\cdot 2, 1\cdot 4)\right]$ (Scalar multiplication)

$= \frac{1}{3}\left[(0,1) + (8,1) + (2,4)\right] = \frac{1}{3}\left[(0+8+2), (1+1+4)\right]$ (Vector addn)

$= \frac{1}{3}\left[(10,6)\right] = \left(\frac{1}{3}\cdot 10, \frac{1}{3}\cdot 6\right) = \left(\frac{10}{3}, 2\right)$

(Scalar multiplication)

b) Determine how to distribute an additional mass of 6gm at the three vertices of the plate to move the balance point of the plate (2,2).

$\boxed{\text{Solution:}}$ Let w_1, w_2 and w_3 denote the masses added at the three vertices, so that $w_1 + w_2 + w_3 = 6$.